

Laser Based Security Lock



EFY LAB

Presented here is a security system that uses an inexpensive laser torch which is usually available with a key ring. The advantage of this security lock is that you can use any laser torch or pointer that is easily available in toy shops. It employs minimal input keys and yet is very secure. It can be used as a door lock, briefcase lock or any other application related to an electronic locking system.

Circuit and working

The block diagram of a laser-based security lock is shown in Fig. 1 and its schematic diagram in Fig. 2. Switches S1 and S2 work as security input keys and switch S3 is used for opening the lock manually. The lock can be an electric strike installed on the door frame to allow access with an access control system or any electronic remote control system.

Electric strikes are generally available in two configurations—fail-secure and fail-safe. In fail-secure configuration, applying electric current to the strike causes it to open; whereas in fail-safe configuration, applying electric current to the strike causes it to lock. A fail-secure configuration is used in this project.

Opening the lock. The use of switch S2, pressing and releasing of switch S1 and supplying the laser pulses at

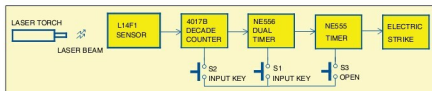


Fig. 1: Block diagram of the laser based security lock

the right time are the security features of this circuit. Switch S2 should not be released during the entire operation, otherwise the counter will get reset and you will not be able to open on the lock.

When switch S1 is pressed, the first in-built timer of IC2 is triggered. Its output pin 5 goes high for four seconds. LED1 glows and transistor T3 conducts. During this time, the collector of T3 becomes low, which in turn pulls the clock enable pin (pin 13) of IC1 to low state. This enables the counter (IC1) to count. During this time, five laser pulses are applied (by you) at photo sensor TP1. These signal pulses go to clock pin 14. LED4 glows and T2 conducts at the fifth pulse. This triggers the in-built second

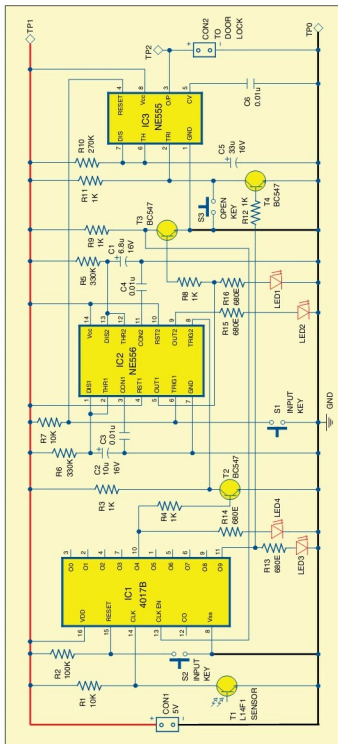


Fig. 2: Circuit diagram of the laser based security lock

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	High when S3 is pressed